

Group 3
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1. How many accidents occurred in NYC, Austin and Chicago

DAMG7370.FINALPROJECT Settings Code Versions

```
29 WHERE l.City IN ('NewYork', 'Austin', 'Chicago')
30 GROUP BY l.City:
31 select * from accident_fact where accident_id like 'NW%'
32
33 1 SELECT
34     l.City AS City,
35     COUNT(DISTINCT a.CRASH_ID_AK) AS cnt_acc
36 FROM DAMG7370.FINALPROJECT.DIM_LOCATION l
37 LEFT JOIN DAMG7370.FINALPROJECT.ACCIDENT_FACT a
38     ON l.Location_ID_SK = a.Location_ID_SK
39 WHERE l.City IN ('NewYork', 'Austin', 'CHICAGO')
40 GROUP BY l.City
41 ORDER BY cnt_acc DESC;
42
43 --2
44 WITH RankedAccidents AS (
```

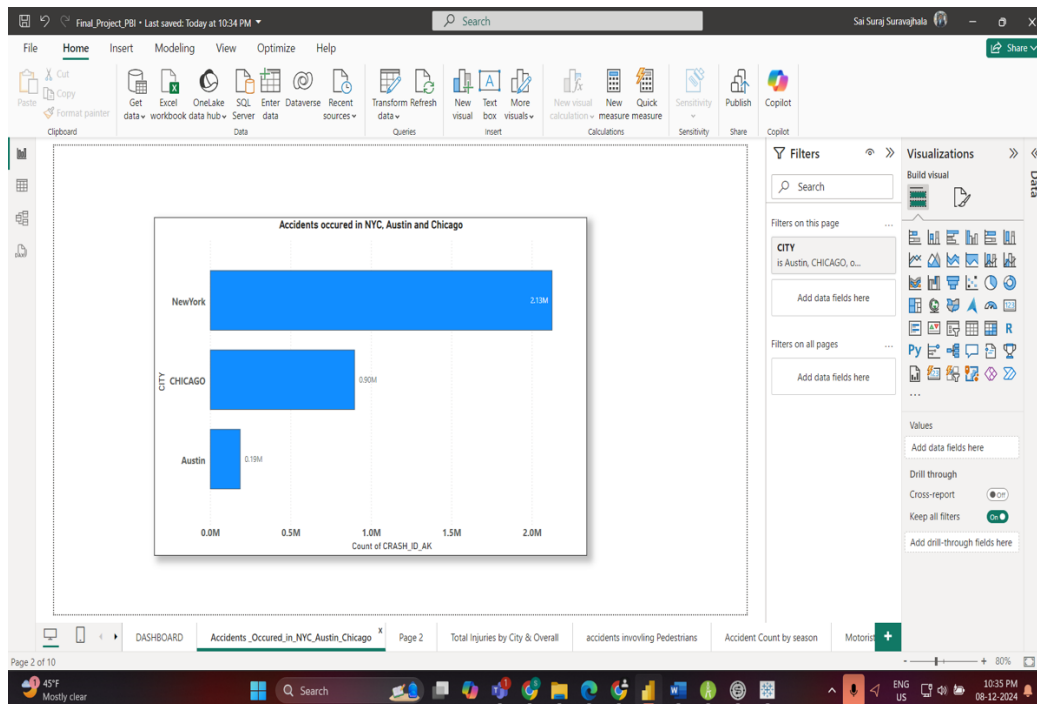
Results Chart

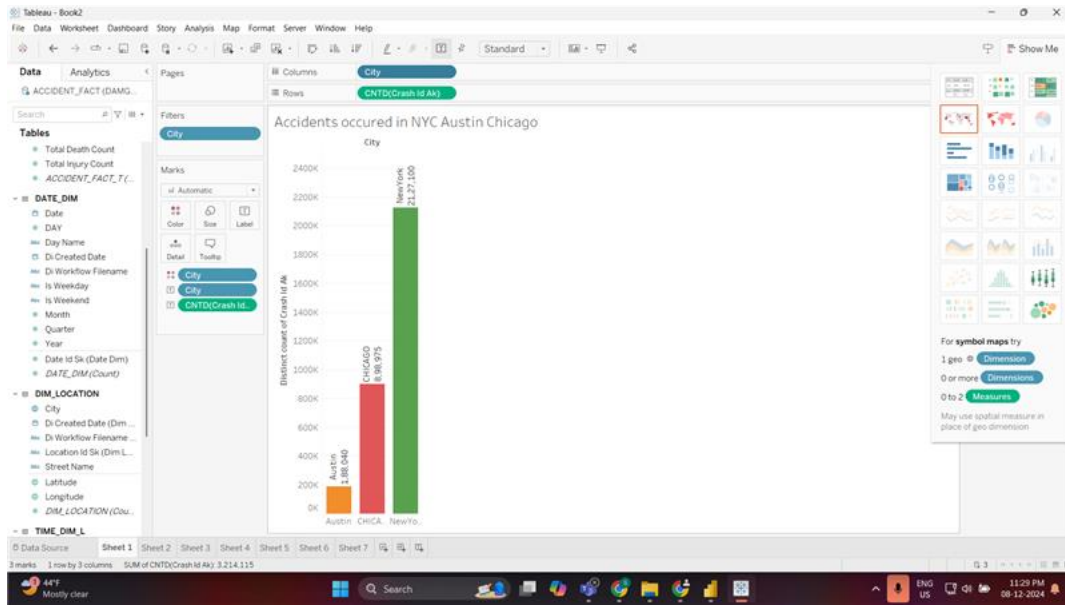
CITY	CNT_ACC
1 NewYork	2127100
2 CHICAGO	898975
3 Austin	188040

Query Details
Query duration 102ms
Rows
Query ID 01b8e723-0000-a2e8-0...

Show more

Ask Copilot





2 Which areas in the cities had the greatest number of accidents

ACCOUNTADMIN COMPUTE_WH (X-Small) Share

DAMG7370.FINALPROJECT Settings Code Versions

```

--2
44 WITH RankedAccidents AS (
45   SELECT
46     L.City,
47     L.Street_Name AS Area,
48     COUNT(distinct a.crash_id_ak) AS Total_Accidents,
49     RANK() OVER (PARTITION BY L.City ORDER BY COUNT(a.crash_id_ak) DESC) AS Rank
50   FROM
51     DAMG7370.FINALPROJECT.ACCIDENT_FACT a
52 )

```

Results Chart

CITY	AREA	TOTAL_ACCIDENTS
1 CHICAGO	WESTERN AVE	24493
2 CHICAGO	PULASKI RD	21685
3 CHICAGO	CICERO AVE	20175
4 NewYork	BROADWAY	17277
5 NewYork	ATLANTIC AVENUE	15322
6 Austin	NOT REPORTED	13831
7 NewYork	BELT PARKWAY	13543
8 Austin	LAMAR	5161
9 Austin	BH 35	4197
10 Montgomery	GEORGIA AVE	1203

Query Details

Query duration 35ms

Rows 12

Query ID 01b8e724-0000-a181-0...

Show more

CITY

CHICAGO 3

NewYork 3

Austin 1

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The screenshot shows the Microsoft Power BI Desktop interface. The main window displays a data table with the following content:

CITY	STREET_NAME	Accident_Count
NewYork	BROADWAY	20372
NewYork	NA	456935
NewYork	WESTERN AVE	1
CHICAGO	BROADWAY	4669
CHICAGO	NA	1538
CHICAGO	WESTERN AVE	24493
Austin	BROADWAY	1
Austin	NA	1

The right-hand pane shows the 'Visualizations' and 'Filters' sections. The 'Visualizations' pane includes a 'Build visual' section with various chart types and a 'Values' section with 'Add data fields here'. The 'Filters' pane includes 'Filters on this page' and 'Filters on all pages' sections, both with 'Add data fields here' buttons. The bottom status bar shows 'Page 3 of 10' and a list of dashboard pages: 'DASHBOARD', 'Accidents_Occurred_in_NYC_Austin_Chicago', 'Page 2', 'Total Injuries by City & Overall', 'accidents involving Pedestrians', 'Accident Count by season', and 'Motorists'.

3 How many accidents resulted in just injuries?

The screenshot shows a SQL query editor with the following query:

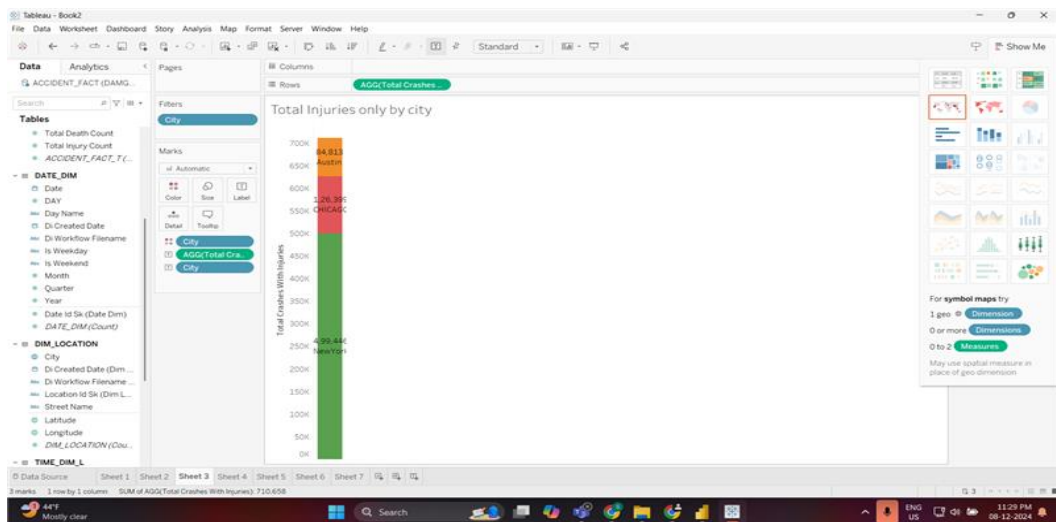
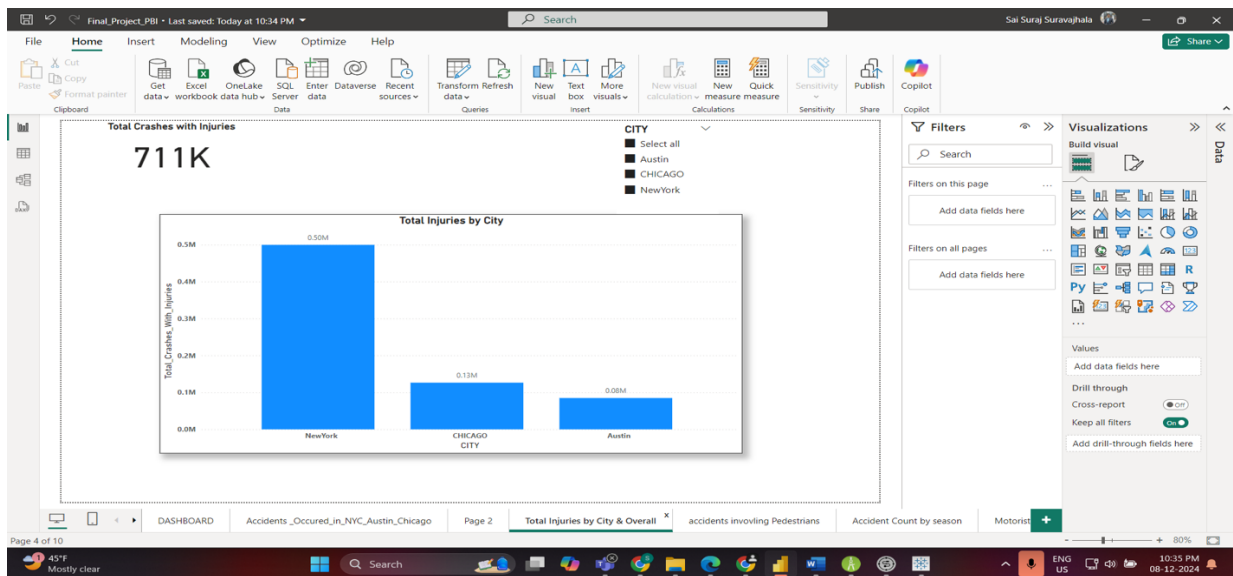
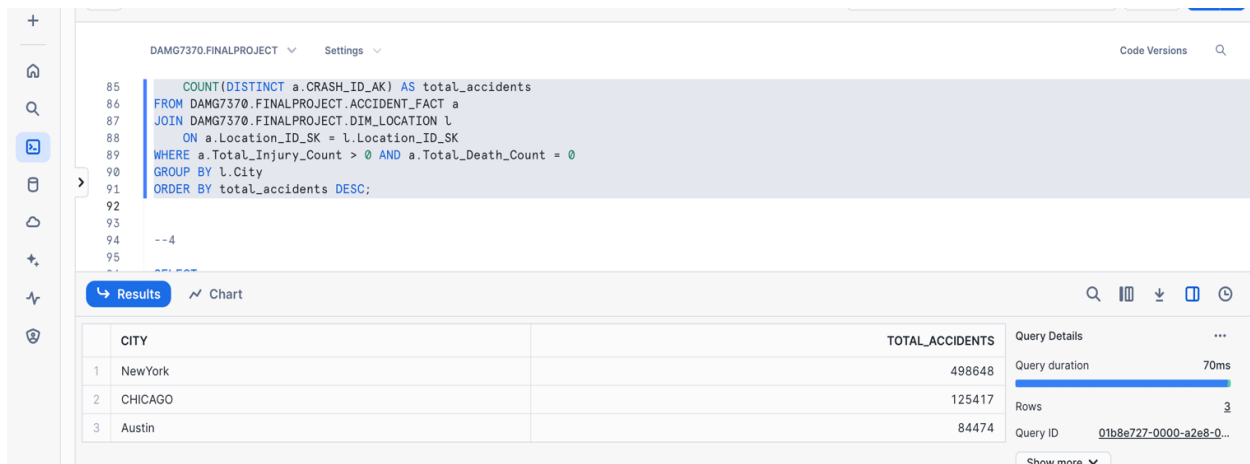
```
--3 Overall
76
77
78 SELECT
79   COUNT(DISTINCT a.CRASH_ID_AK) AS total_accidents
80 FROM DAMG7370.FINALPROJECT.ACCIDENT_FACT a
81 WHERE a.Total_Injury_Count > 0 and total_death_count= 0;
82
83 --BY City
84 SELECT
85   l.City AS city,
86   COUNT(DISTINCT a.CRASH_ID_AK) AS total_accidents
87 FROM DAMG7370.FINALPROJECT.ACCIDENT_FACT a
```

The 'Results' pane shows the following data:

TOTAL_ACCIDENTS
710864

The 'Query Details' pane shows the following information:

- Query duration: 32ms
- Rows: 1
- Query ID: 01b8e727-0000-a2e8-0...



4 How often are pedestrians involved in accidents?

The screenshot shows a data query interface. At the top, there's a navigation bar with various icons and a search bar. Below that, a query editor displays a SQL query:

```
--4
SELECT
  l.city AS city,
  COUNT(DISTINCT fa.crash_id_ak) AS pedestrian_involved
FROM
  accident_fact fa
JOIN
  dim_location l
ON
  fa.location_id_sk = l.location_id_sk
```

Below the query editor, the results are displayed in a table:

CITY	PEDESTRIAN_INVOLVED
1 CHICAGO	21312

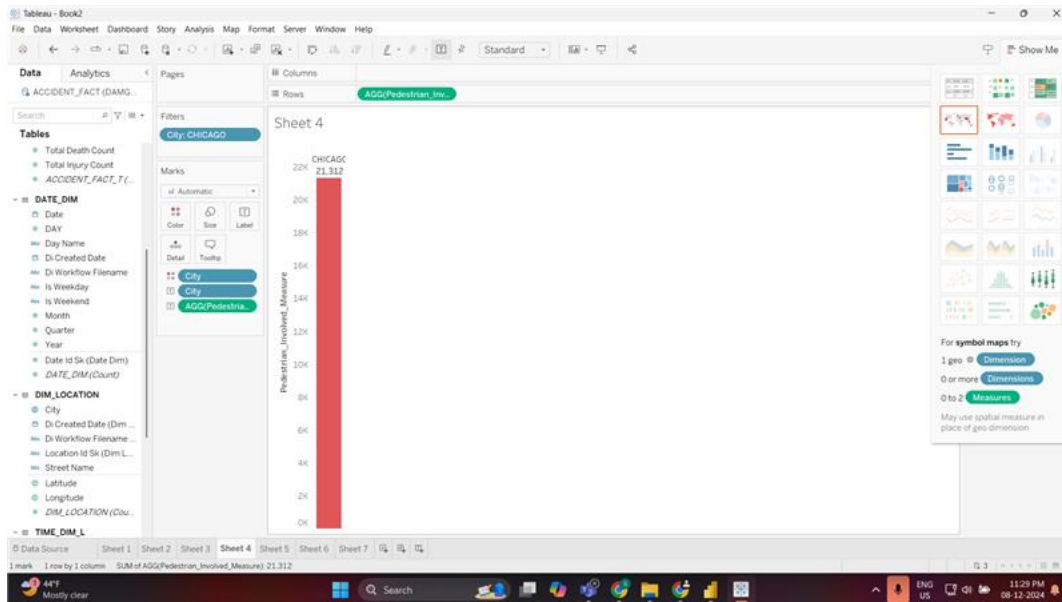
On the right side of the results table, there's a 'Query Details' panel showing:

- Query duration: 44ms
- Rows: 1
- Query ID: 01b8e728-0000-a1bd-0...

The screenshot shows a Power BI dashboard. The main visual is a bar chart titled 'Pedestrian_Involved_measure by CITY'. The Y-axis is labeled 'Pedestrian_Involved_measure' and ranges from 0K to 20K. The X-axis is labeled 'CITY' and has a single bar for 'CHICAGO' with a value of 21.3K. The chart is displayed in a 'Card' visual type.

On the right side, there's a 'Visualizations' pane with various chart options. Below it, there's a 'Filters' pane with options to add data fields here. At the bottom, there's a 'Values' section with options to add data fields here, a 'Drill through' section, and a 'Cross-report' section with a toggle switch.

The dashboard is titled 'Final_Project_PBI' and was last saved at 10:34 PM. The user is Sai Suraj Suravajhala. The dashboard includes a navigation bar with various icons and a search bar. The bottom of the dashboard shows a 'Page 5 of 10' indicator and a 'DASHBOARD' tab.



5 When do most accidents happen (In which season)?

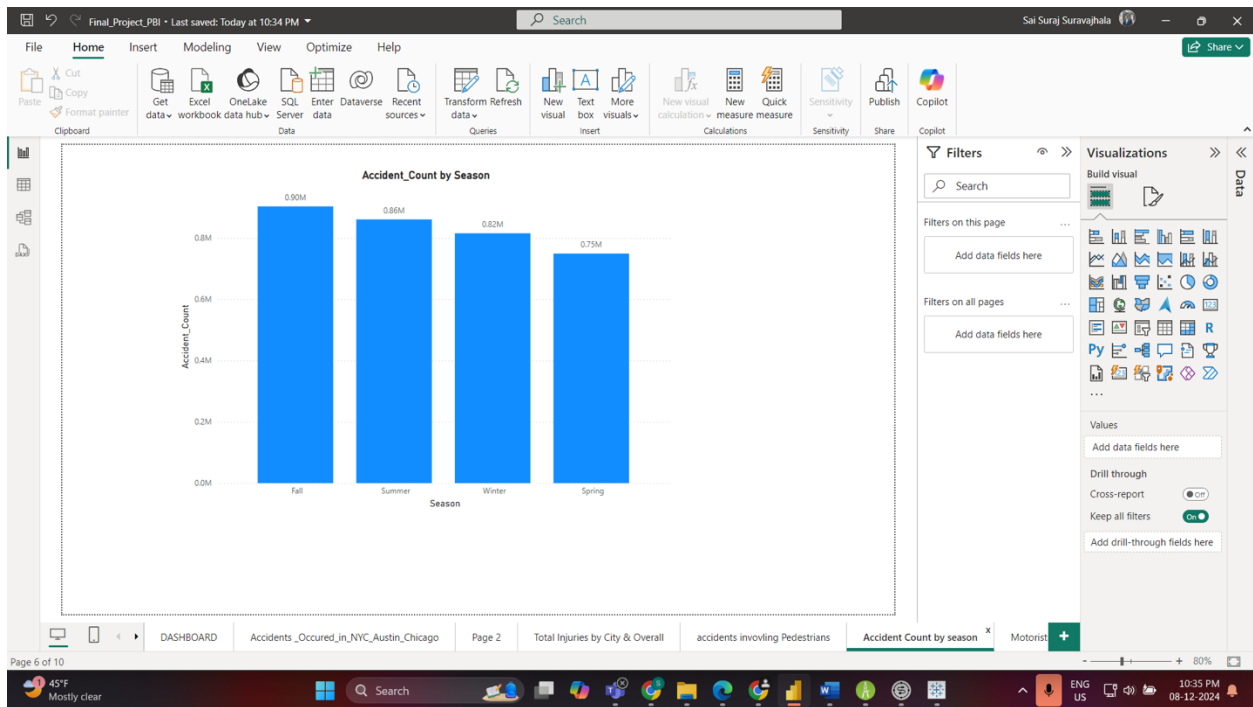
```

112 SELECT SUM(ACCIDENT_COUNT) AS ACCIDENT_COUNT
113 FROM ACCIDENT_FACT
114 WHERE d.month IN (11, 12, 1)
115
116 SELECT
117 CASE
118 WHEN d.month IN (11, 12, 1) THEN 'Winter'
119 WHEN d.month IN (3, 4, 2) THEN 'Spring'
120 WHEN d.month IN (6, 7, 5) THEN 'Summer'
121 WHEN d.month IN (9, 10, 8) THEN 'Fall'
122 ELSE 'Unknown'

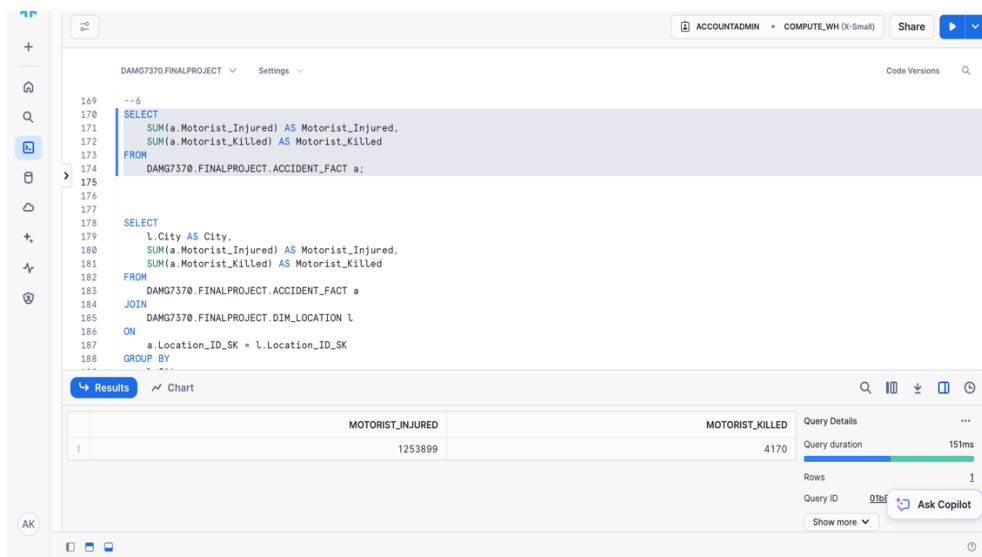
```

SEASON	ACCIDENT_COUNT	ACCIDENT_COUNT
1 Fall	903799	749631
2 Summer	861277	
3 Winter	815844	
4 Spring	749631	

Ask Copilot



6 How many motorists are injured or killed in accidents?



Browser tabs: Gmail, YouTube, Maps, News. Account: ACCOUNTADMIN, Compute: COMPUTE_WH (X-Small). Query ID: 01b8e72c-0000-a0b6-9...

```
1/4
175 DAMG7370.FINALPROJECT.ACCIDENT_FACT a;
176
177
178 SELECT
179   L.City AS City,
180   SUM(a.Motorist_Injured) AS Motorist_Injured,
181   SUM(a.Motorist_Killed) AS Motorist_Killed
182 FROM
183   DAMG7370.FINALPROJECT.ACCIDENT_FACT a
184 JOIN
185   DAMG7370.FINALPROJECT.DIM_LOCATION L
```

CITY	MOTORIST_INJURED	MOTORIST_KILLED
1 NewYork	1242115	3460
2 Austin	5275	676
3 CHICAGO	666	0
4 Montgomery	0	0

Query Details: Query duration 428ms, Rows 4. Ask Copilot button.

Power BI Desktop: Final_Project_PBI - Last saved: Today at 10:34 PM. User: Sai Suraj Suravajhala.

Motorists Killed and Injured by City

Sum of MOTORIST_KILLED and Sum of MOTORIST_INJURED

CITY	Sum of MOTORIST_KILLED	Sum of MOTORIST_INJURED
NewYork	3460	1242115
Austin	676	5275
CHICAGO	0	666
Montgomery	0	0

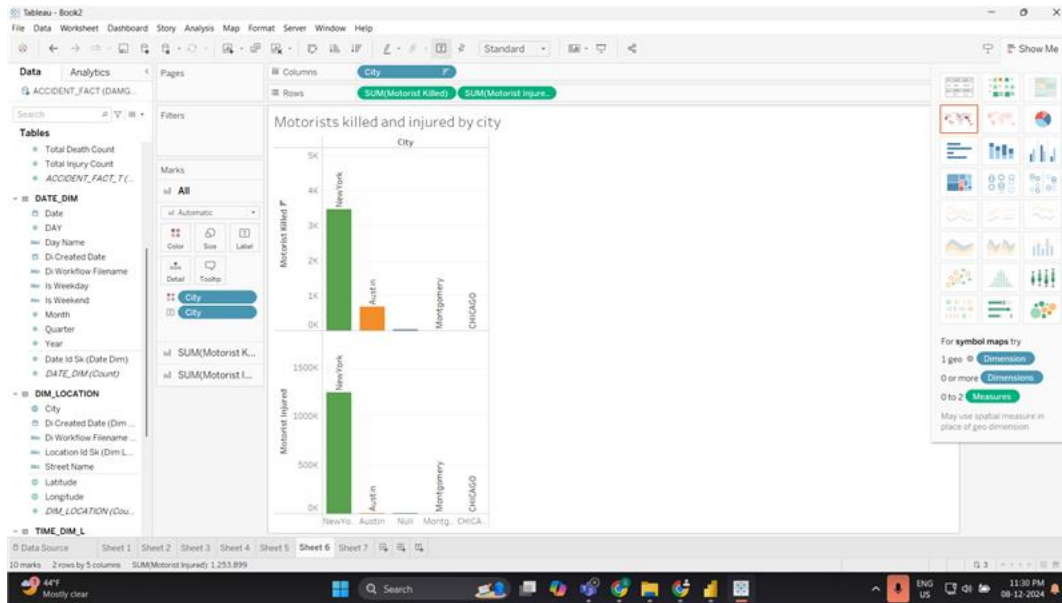
Motorists Killed

4.17K (0.00K to 8.34K)

Motorists Injured

1.25M (0.00M to 2.51M)

Visualizations: Bar chart, Gauge charts. Filters: City & Overall, accidents involving Pedestrians, Accident Count by season, Motorists Killed and Injured by City, Time based Analysis of accidents, 7th, Fatality Analysis.



7 Which top 5 areas in the cities have the most fatal number of accidents?

6:45pm 2024-12-05 7:02pm 2024-12-05 7:06pm 2024-12-05 7:10pm Load sample data with Pyt... 2024-12-05 1:40pm

ACCOUNTADMIN COMPUTE_WH (X-Small) Share

DAMG7370.FINALPROJECT Settings Code Versions

```

196 -----
197 --7
198
199 WITH RankedFatalAccidents AS (
200     SELECT
201         l.City,
202         l.Street_Name AS Area,
203         COUNT(a.Accident_ID) AS Fatal_Accidents,
204         RANK() OVER (PARTITION BY l.City ORDER BY COUNT(a.Accident_ID) DESC) AS Rank
205     FROM
206         DAMG7370.FINALPROJECT.ACCIDENT_FACT a
  
```

Results Chart

CITY	AREA	FATAL_ACCIDENTS
1 Austin	NOT REPORTED	90
2 NewYork	BROADWAY	74
3 NewYork	3 AVENUE	73
4 NewYork	BELT PARKWAY	65
5 NewYork	ATLANTIC AVENUE	60
6 NewYork	LINDEN BOULEVARD	57
7 CHICAGO	PULASKI RD	32
8 Austin	LAMAR	31
9 CHICAGO	HALSTED ST	31

Query Details

- Query duration: 361ms
- Rows: 15
- Query ID: 01b8e72c-0000-a1bd-0...

City: Austin, NewYork, CHICAGO

8 Time-based analysis of accidents

- Time of the day, day of the week, weekdays, or weekends

The screenshot shows a SQL query editor with a query that joins accident facts with a time dimension and counts them by time period. The query is as follows:

```
241 DAMG7370.FINALPROJECT.ACCIDENT_FACT a
242 JOIN
243 DAMG7370.FINALPROJECT.TIME_DIM t
244 ON
245 a.TIME_ID_SK = t.TIME_ID_SK
246 GROUP BY
247 t.Time_Period;
248
249 --Count By day
250
251
252 SELECT
```

The results table shows the following data:

TIME_PERIOD	COUNT_ACC
1 Morning	1690421
2 Afternoon	273422
3 Late Night	261317
4 Early Morning	1105391

Query Details: Query duration 678ms, Rows 4, Query ID 01b8e72d-0000-a0b6...

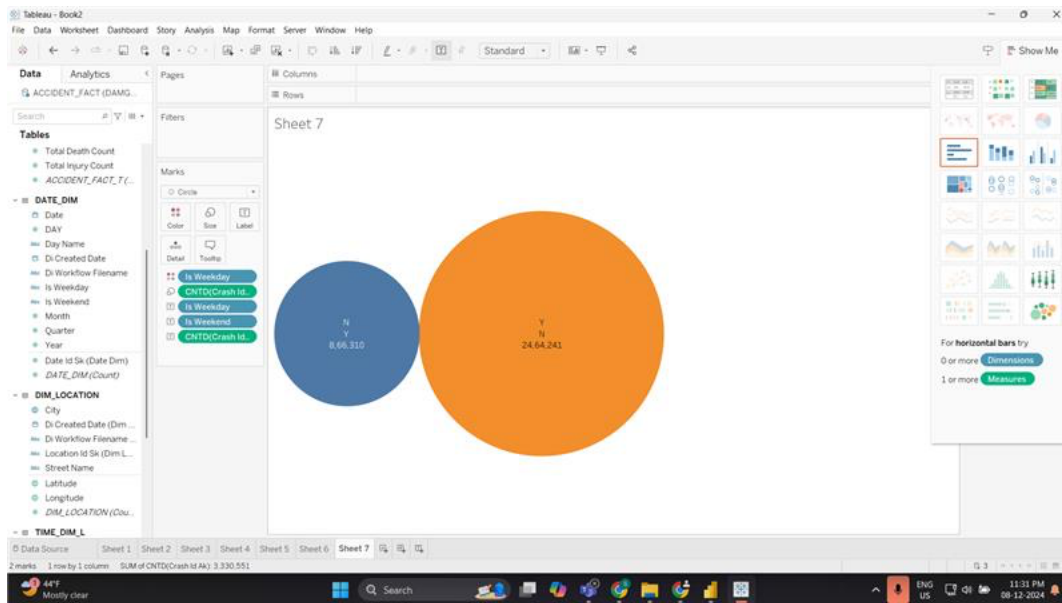
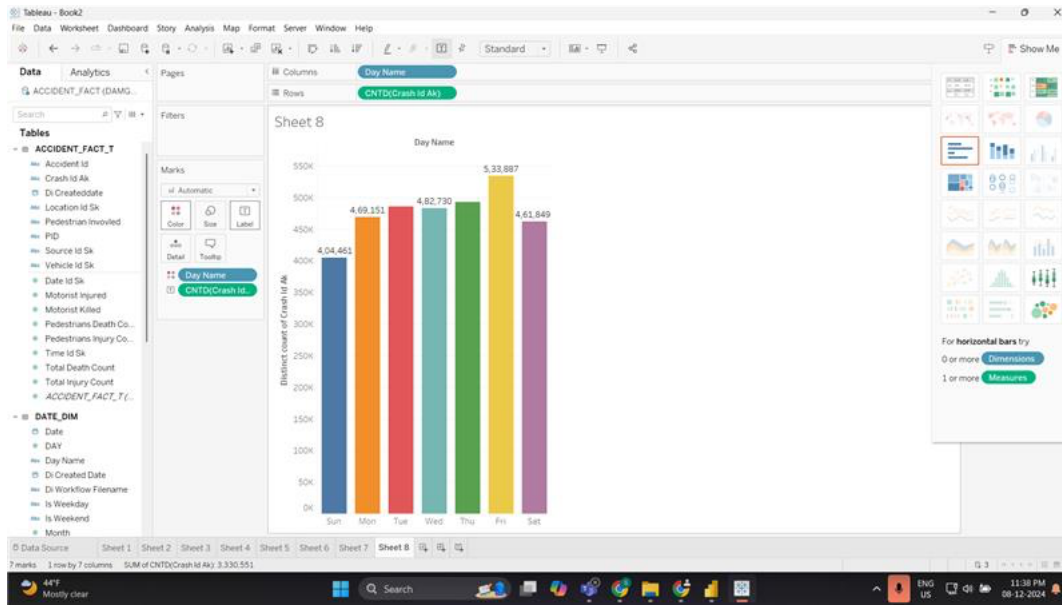
The screenshot shows a SQL query editor with a query that counts distinct crash IDs by day of the week. The query is as follows:

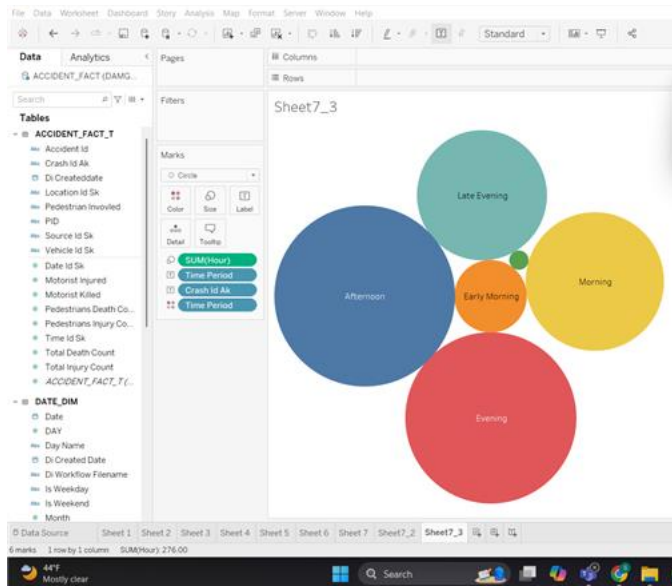
```
254 COUNT(DISTINCT a.CRASH_ID_AK) AS Count_Acc
255 FROM
256 DAMG7370.FINALPROJECT.ACCIDENT_FACT a
257 JOIN
258 DAMG7370.FINALPROJECT.DATE_DIM d
259 ON
260 a.Date_ID_SK = d.Date_ID_SK
261 GROUP BY
262 d.day_name;
263
264
265
```

The results table shows the following data:

DAY	COUNT_ACC
1 Thu	492769
2 Wed	482730
3 Sun	404461
4 Mon	469151
5 Sat	461849
6 Tue	485704
7 Fri	533887

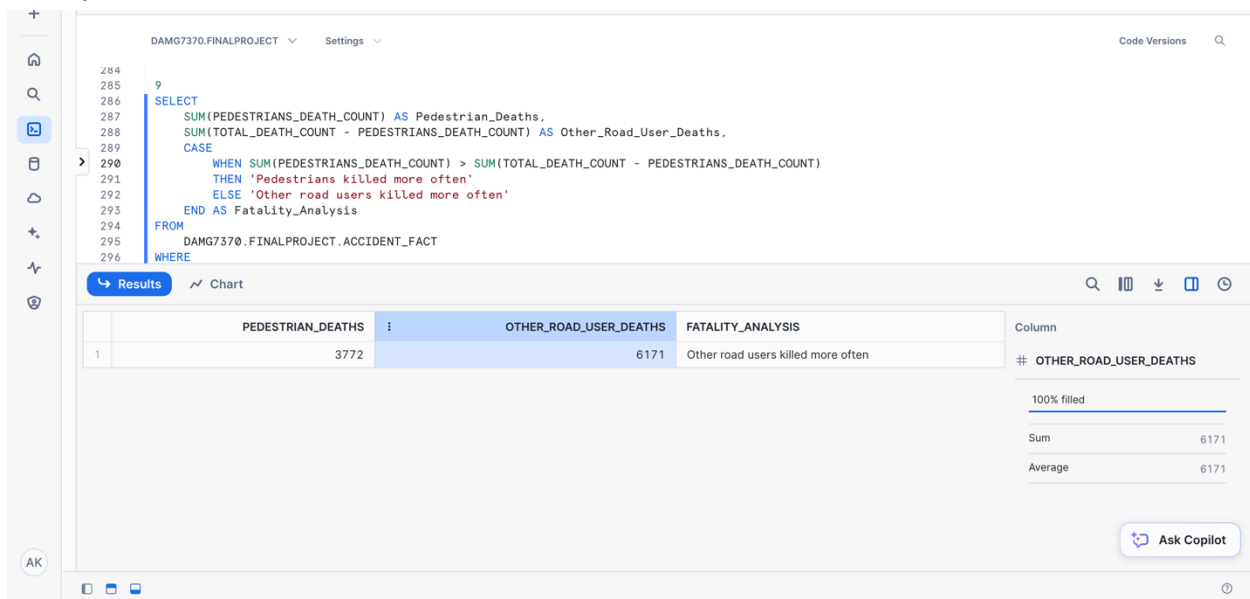
Query Details: Query duration 707ms, Rows 7, Query ID 01b8e72d-0000-a334...

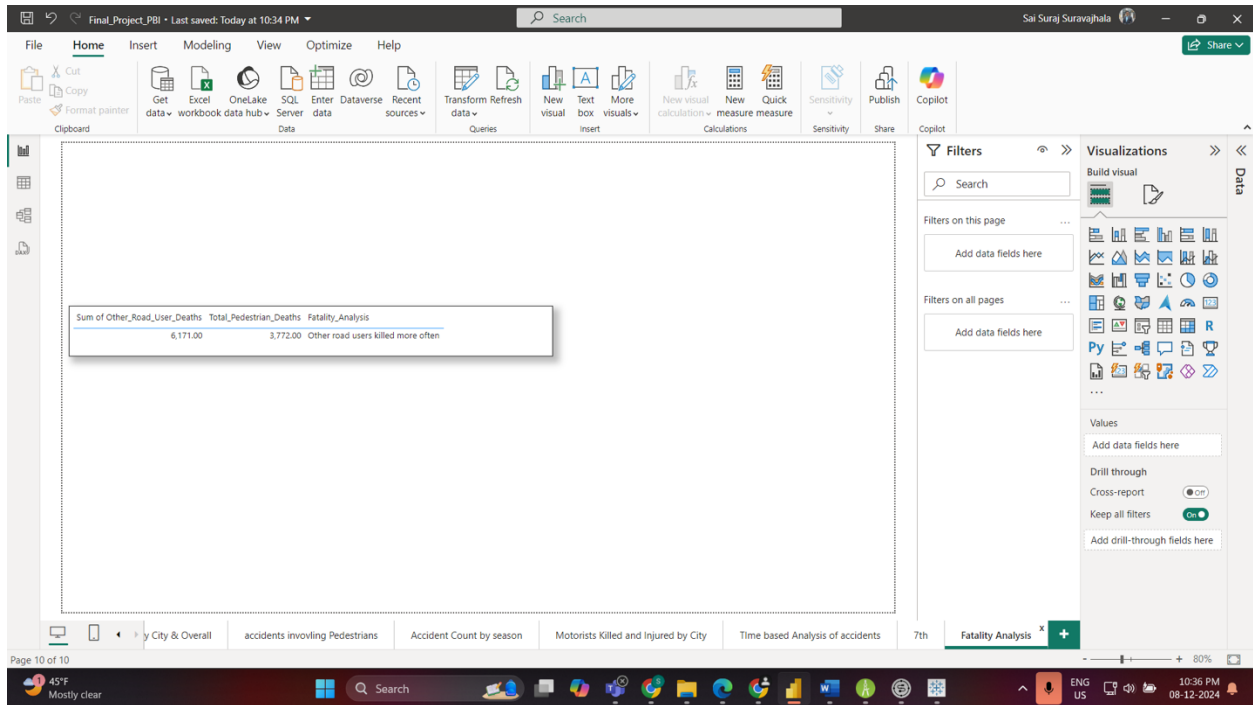




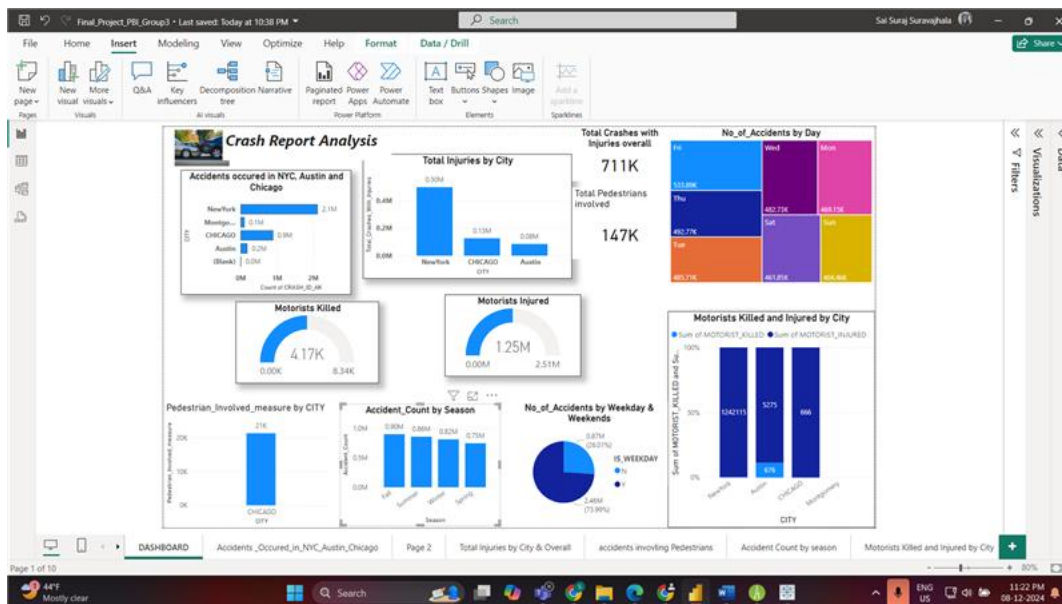
9 Fatality analysis

- Are pedestrians killed more often than road users?





Dashboard of Power BI



The screenshot shows the Tableau Desktop interface. The main view is a map of the United States, titled 'Sheet 5'. The map displays data points for 'AVG(Longitude)' and 'AVG(Latitude)'. The left sidebar shows the data source 'ACCIDENT_FACT (DAMG...)' and the dimensions 'DATE_DIM' and 'DIM_LOCATION'. The right sidebar shows a 'Symbol maps try' section with options for 'Dimension' and 'Measure'.